

Lecture 5 Quiz Questions and Answers

1 Select the correct statement/s

proportional controller is an amplifier with adjustable gain

☐ 0%

derivative controller is as an amplifier with a "knob" that can be adjusted up or down.

☐ 0%

goal of derivative feedback is to improve closed-loop system stability

☐ 10%

1 and 2

☐ 26%

1 and 3 ☒

☐

2 Derivative control is almost never used by itself, because:

it reduces the transient response

☐ 0%

it does not supply information on the desired end state ☒

☐ 0%

it tends to provide no response to noise

☐ 0%

1 and 2

☐ 0%

2 and 3

☐ 0%

3

Goal of integral control is to

minimize the steady-state tracking error

☐ 0%

minimize steady-state output response to disturbances

☐ 0%

use as a reset control

☐ 0%

1 and 2

☐ 0%

all of the above ☒

☒ 0%

4

Select correct statement/s

Derivative action is used to speed the effect of the proportional action ☒

☒ 0%

proportional action doesn't add a steady state error

☐ 0%

response of a second-order system to a unit step input isn't dependent on the damping

☐ 0%

1 and 2

☐ 0%

none of the above

☐ 0%

5

Select correct statement/s

Proportional control of a first-order system gives a large error for low gain ✓

☒ 0%

infinite proportional gain causes system to respond as a unity feedback

☐ 0%

high proportional gain increase the time constant

☐ 0%

all of the above

☐ 0%

none of the above

☐ 0%